Here are the steps to implement the image upload feature using private storage:

9.8.4.1 Step 1: Create a New Laravel Project

We will create a new Laravel project for this lab. Open your terminal and run:

```
laravel new image-upload-private
cd image-upload-private
code .
```

You should see the Laravel project in your editor.

9.8.4.2 Step 2: Set Storage to Local (Private)

Open the `.env` file and set the `FILESYSTEM_DISK` to `local`. This will ensure that files are stored in the private storage directory (`storage/app`).

```
FILESYSTEM_DISK=local
```

9.8.4.3 Step 3: Create the Controller

We will create a controller to handle the image upload logic. Run the following command in your terminal:

```
php artisan make:controller PrivateImageUploadController
```

You can find the controller in `app/Http/Controllers/PrivateImageUploadController.php`. This controller will handle the form display, image upload, and result display.

Edit `app/Http/Controllers/PrivateImageUploadController.php`:

```
<?php
namespace App\Http\Controllers;

use Illuminate\Http\Request;
use Illuminate\Support\Facades\Storage;
use Symfony\Component\HttpFoundation\Response;
use Illuminate\Support\Str;

class PrivateImageUploadController extends Controller
{
    public function showForm()
    {
        return view('upload');
    }

    public function handleUpload(Request $request)
```